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SURVEY

MLOps Components, Tools, Process, and Metrics: A Systematic Literature Review

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ABSTRACT With the growing popularity of machine learning, implementations of the environment for developing and maintaining these models, called MLOps, are becoming more common. The number of publications in this area is relatively small, although growing rapidly. Our goal was to review the current state of the literature in the MLOps area and answer the following research questions: What classes of tools are used in MLOps environments? Which tool implementations are the most popular? What processes are implemented within MLOps? What metrics are used to measure the effectiveness of MLOps implementation? Based on this review, we identified classes of tools included in the MLOps architecture, along with their most popular implementations. While some tools originate from DevOps practices, others, such as Model Orchestrators, Feature Stores, and Model Repositories, are unique to MLOps. We propose a reference MLOps architecture based on these findings and outline the stages of the model production process. We also sought metrics that would allow us to assess and compare the effectiveness of MLOps practices, but unfortunately, we were unable to find a satisfactory answer in this area.

INDEX TERMS Architecture, machine learning operations, MLOps.

I. INTRODUCTION

The growing popularity of Machine Learning and Artificial Intelligence increases the amount of work involved in implementing, maintaining and developing models, as well as ensuring their efficiency and reliability. To address this growing workload, specialized environments have emerged to streamline the operationalization of machine learning models. These environments, along with the processes of maintaining and developing models, are collectively called MLOps (Machine Learning Operations). This acronym comes from a combination of machine learning and DevOps, which refers to operations around software development (Development Operations). Although the literature offers no unified definition of MLOps, this study references existing perspectives to provide context. However, we will quote some definitions close to us to explain the meaning of this word. Heydari et al. [1] presents the following definition "MLOps stands for Machine Learning Operations, is a discipline that focuses on the continuous delivery cycle of machine learning models through automated pipelines". John et al. [2] add the aspect of best practices and concepts in their definition: "MLOps includes best practices, concepts and a development culture to accelerate the operationalisation of ML models by bridging the gap between the development (Dev) and operations (Ops) team". Adding to these perspectives, Wikipedia emphasizes reliability and efficiency in defining MLOps as 'a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently'. These three definitions complement each other and provide a complete picture of what MLOps means.

The AI market worth in 2023 was 196,6B$ and it is projected to grow at a very rapid rate of 36,6% yearly (CAGR) from 2024 to 2030 [50]. As the AI market grows, so does MLOps. The trend of its popularity is depicted in Fig. 1, which presents the percentage of MLOps searches over years. Therefore, MLOps is an important market area and worth deep analysis. This study aims to identify solutions that facilitate the implementation of MLOps in organizations. From this goal, we derived the following research questions:

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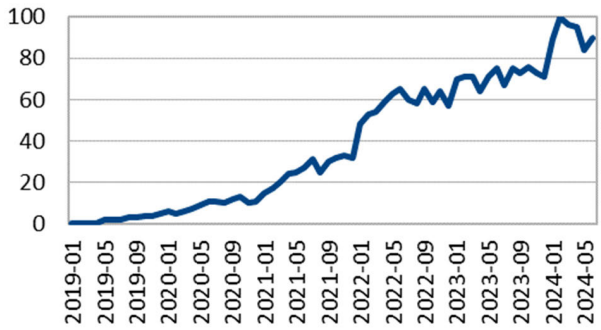

FIGURE 1. MLOps search percentage, Data source: Google trends.

RQ1. What classes of tools do MLOps solutions comprise?

RQ2. What tools are used to implement MLOps in organizations?

RQ3. What processes are included in MLOps solutions?

RQ4. What metrics are used to assess the success of a MLOps implementation?

In this study, the term 'MLOps solution' refers to a comprehensive framework of tools and processes designed to support the development and deployment of ML models within organizations. We use the term 'component' as a class of tools. We also define 'tool' as one of the numerous implementations of a component. Thus, each component can have various implementations as different tools from different vendors. The MLOps process refers to a set of activities performed by data scientists or fully automated activities. These activities involve the use of MLOps tools, but they are not synonymous.

The remainder of this paper is organized as follows. Section II discusses the literature review and the rationale for conducting this review. In Section III, you can find a detailed explanation of the steps taken to prepare the review and its limitations. Section IV provides answers to the research questions. The paper's summary and plans for future work are outlined in Section V.

II. RELATED LITERATURE

The existing literature includes several reviews related to MLOps. We have found five reviews related to ours. First, the review [3] examines MLOps tools and their features. This is an area that we also explore in this review. However, we would like to present the current results for 2024, whereas in the article in question the latest references are from 2020. In addition, its authors used non-peer-reviewed online sources, which is justified by the early stage of MLOps development in 2020 and the limited number of publications and tools in this area. Review [4], which includes articles published up to 2022, leverages peer-reviewed sources such as IEEE Xplore, ACM, Springer, and Elsevier. However, it was limited to 18 articles discussed, which is also justified by the stage of development of the MLOps area. The other literature reviews have a similar limitation and we will not raise this argument when discussing subsequent reviews. The review [4] discussed how MLOps is understood in the literature, how it differs from DevOps, and what the most important topics are discussed in it. However, we want to focus on the architectural dimension of MLOps and to answer other research questions. The next review we discuss is [5]. The authors answer questions about what the MLOps literature focuses on in Data Science projects, what activities it automates, and what the applications of MLOps are. The authors also present a high-level categorization of articles into areas such as tool, review, framework and application. Therefore, they focus more on the management area than on the architectural area. Ask Kolltveit et al. [6] identified ways of operationalizing models: packaging and integration (containerization), deployment, serving and inference, monitoring and logging. They describe challenges during those steps and tools used during operationalization of models. The review [7] is the most comprehensive among those discussed. The questions answered by the authors concern three issues: terms used to describe AI production processes, tasks included in these processes, and challenges related to them. Based on our literature analysis, we did not find an up-to-date review that answered our research questions.

III. RESEARCH METHOD

A. RESEARCH QUERY

When working on the literature review, we followed the guidelines by Kitchenham and Charters [8]. We started the literature review by selecting the databases that were searched. We have selected the most popular databases that have been used in other reviews, namely:

A. EBSCO
B. IEEE
C. Proquest
D. Science Direct
E. Springer

We performed the following query on these databases: mlops OR "machine learning operations" OR "ML operations" OR "AI operations" OR "Artificial intelligence operations".

Each database had different detailed search options related to the query. Therefore, we have selected the following search options in individual databases:

- EBSCO: We excluded the following databases: GreenFILE, Health Source, MEDLINE, Agricola.
- Science Direct: We have filtered the results to include the following subject areas: Computer Science; Engineering; Decision Sciences; Business, Management and Accounting.
- IEEE: we did not filter the results.
- ProQuest: We filtered results to include Scholarly Journals.
- Springer: We selected the following filters: Language – English; Disciplines – Computer Science, Engineering, Business and management; Subdisciplines – Artificial intelligence, Computational intelligence, Machine learning, Computer communication networks, Computer applications, Computer engineering, networks.

B. SELECTION PROCESS

From the results obtained in response to the query, we selected texts that were included in the literature review. The selection of articles comprised two stages. The first involved selecting articles based on titles and abstracts. At this stage, we excluded articles clearly unrelated to our field and retained those potentially relevant. In the second stage, the content of the articles was analyzed to select the final set from those selected in the first stage. To make the selection process clear and repeatable, we always followed the inclusion and exclusion criteria. We included publications that met the following criteria:

- describe the components of MLOps,
- describe MLOps processes,
- describe tools used to implement MLOps,
- describe metrics that allow for assessing the effects of MLOPS implementation.

We excluded publications that met the following criteria:

- present a technical tutorial of a single tool,
- focus on Auto ML processes without touching on the MLOps area,
- present an implementation of models whose production use they call MLOps,
- present notation (UML or similar) for modeling MLOps,
- present network solutions that can support the work of ML models.

For each article that passed the second phase, we recorded for statistical purposes:

- what classes of tools are used in MLOps architecture design,
- what solutions are used that implement those classes of tools,
- what steps are proposed in MLOps processes,
- what KPIs were used to assess the effects of MLOps implementation.

The selection of publications began with 2615 search results. In the first stage, publications that would definitely not be included in the review based on the title and abstract were eliminated. 135 publications passed the first stage. Finally, 41 publications were selected for the review after the second stage. The distribution of articles by stages and databases is presented in Figure 2, which shows that not all text databases gave equally satisfactory results. Although Springer produced the highest number of initial results, none of its publications met the inclusion criteria.

The IEEE database was characterized by the highest efficiency, as 8.5% of the articles were used in the review. The exact percentage distribution is shown in Fig. 3. Fig. 4 shows the division of publications by years, and clarifies that the MLOPS area gains popularity. Because database queries were made in April 2024, the result for 2024 was low.

C. LIMITATIONS

In our work, we found two significant limitations that may affect the results. The first one is the data sources. MLOps is of interest not only to scientists but also to private companies.

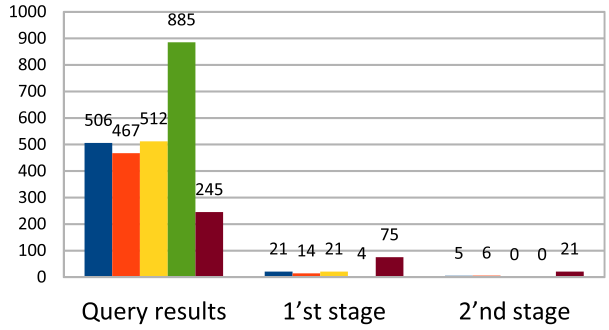

FIGURE 2. Number of publications in each stage.

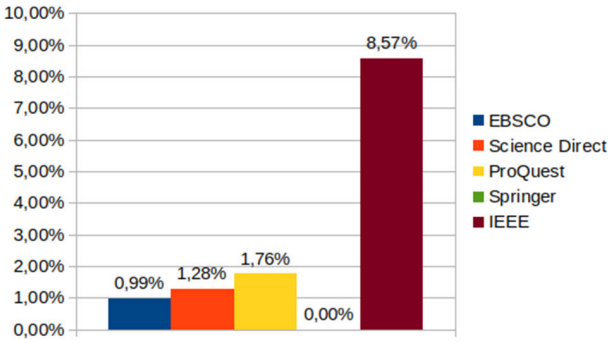

FIGURE 3. Percentage of query results passed into 2'nd stage.

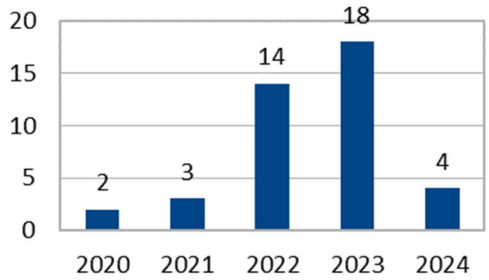

FIGURE 4. Number of publications in years after 2'nd stage.

We assume that a lot of work in this area has been done in organizations that do not publish their results. The second limitation is the short time of MLOps development and its rapid evolution at the same time. At the moment, it is not possible to clearly define the MLOps architecture for years and at the same time it is necessary to update the results frequently.

IV. RESULTS

A. AD RQ1: CLASSES OF MLOps TOOLS

Upon analyzing the texts, we noticed that some classes of tools are more popular than others. The most common component was the Model Repository, which we found in 22 publications. This is an architectural element responsible for storing files with trained models, as well as metadata about these models. This metadata typically includes

TABLE 1. References to classes of MLOps tools in literature.

Component	Related Papers	Total num.
Model Repository	[9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]	22
Model Orchestrator	[9], [10], [12], [31], [32], [33], [15], [16], [17], [34], [20], [35], [21], [22], [23], [24], [25], [27], [28], [36],	20
CI/CD	[37], [11], [12], [15], [16], [18], [19], [34], [20], [21], [22], [23], [24], [38], [26], [39], [29], [36], [30]	19
Feature Store	[12], [14], [15], [20], [21], [22], [23], [24], [26], [27], [28], [36]	12
Database	[9], [10], [11], [21], [22], [23], [25], [27], [36]	9
Model Monitoring	[14], [15], [17], [19], [35], [21], [22], [23], [25], [36], [30]	11
Compute Spread	[37], [40], [29]	3
Code Repository	[37], [12], [15], [16], [18], [19], [34], [20], [35], [23], [24], [26], [41], [39], [36], [30]	16
Containerization	[11], [12], [33], [16], [17], [34], [35], [22], [25], [27], [41], [28], [29], [36], [30]	15
Container Repository	[37], [9], [21], [24], [25], [26], [29], [36]	8
Container Orchestration	[9], [12], [15], [17], [18], [34], [35], [22], [24], [26], [27], [41]	12
Queue Management	[17],[18]	2

measures of model quality. Depending on the model type, different metrics may be stored.

The next most popular tool types were the Model Orchestrator (20 occurrences) and CI/CD (19 occurrences). These tools may seem similar to each other, but they have different responsibilities in the MLOps architecture. The Model Orchestrator handles the processes of training, monitoring, and scoring models. The CI/CD tool was used to build containers that support processes executed in the Model Orchestrator. Additionally, CI/CD tools were used in software testing and implementation. This especially affected those models that were exposed by the REST API, because then the model was "wrapped" in a REST framework, such as DjangoRestFramework or Flask. What both classes of tools have in common is the ability to execute pipelines. Therefore, it was possible to find a proposal for CI/CD class tools to act as a Model orchestrator (such as Jenkins). However, such situations were rare, and most often the tools belonging to both groups were separate.

Another group of tool classes found in MLOps solutions are classic DevOps tools (which also include CI/CD tools). We can include such classes as: Code Repository, Containerization, Container Repository, Container Orchestration and Queue Management. Although these tools are not explicitly mentioned in all publications, we can suspect that they are included in most solutions in the MLOps area. Using a Code Repository and Containerization along with tools to manage containerization (repository and orchestrator) is a modern standard of software development, and this also applies to software in Machine Learning. Using a queue management system is not standard, but is used to perform tasks that take longer than would be acceptable with the REST API. This applies in particular to scoring tasks, as training is most often managed by the Model Orchestrator.

The next classes of tools are Feature Store (12 occurrences) and Database (9 occurrences). Both classes play a similar role in the MLOps architecture. However, they are often combined to get the best effect. Databases store data that is then used in MLOps processes. A Feature Store differs from a database by emphasizing data access and organization specifically for machine learning purposes. Using it, you can organize data as features. Data organized in this way can then be downloaded from the database using a special API provided by the Feature Store. Most often, this is done using a ready-made Python library. Some of the Feature Store class solutions are equipped with their own database and make up a complete data access and storage solution. Others focus on the aspect of data access and management, but require providing them with a database on which they can operate. During the review, we encountered every combination:

- database without a feature store,
- stand-alone feature store,
- feature store with an external database.

Another class of tools in the MLOps architecture is Model Monitoring (11 occurrences). They monitor the quality of models. This does not just apply to storing model metrics, which is often done by a model repository. Here, it is primarily about calculating these metrics, as well as other auxiliary indicators, thanks to which you can improve the model in advance, before the results deteriorate significantly. This refers to indicators such as data drift.

The last class of tools we have encountered is Compute Spread. Although we found it in only 3 publications. Its task is to distribute computations between servers, but at a lower level than a Model Orchestrator does. Model Orchestrator operates at the level of individual steps in the pipeline. Compute Spread Server concerns the spreading of calculations in a single step, such as the spreading of calculations in training a neural network.

Table 1 summarizes the occurrences of each tool class in the reviewed publications, indicating where they were mentioned.

B. AD RQ2: IMPLEMENTATIONS MLOps TOOLS

There are numerous tools on the market that fall into the previously mentioned classes. We have identified 34 tools in this section. Table 2 shows all tools and literature references. We included only cases where authors explicitly reported selecting a tool for use, rather than merely mentioning its existence. So, for example, the HopsWorks tool was not listed as Feature Store, although various authors indicated that they were considering using it (but chose another). The total

TABLE 2. References to MLOps tools in literature.

Class	Tool	Related Papers	Total num.
Feature Store	Feast	[14]	1
Database	PostgreSQL	[10], [22], [36]	3
	GraphDB	[10]	1
Model Repository	MLflow	[9], [13], [14], [17], [18], [19], [27], [36]	8
	Azure ML	[11]	1
Code Repository	Github	[37], [19], [39]	3
	Gitlab	[34]	1
CI/CD	Github Actions	[37], [19]	2
	Jenkins	[33], [26], [39], [36]	4
	Gitlab CI	[29]	1
	Drone	[34]	1
Compute Spread	Flower	[37]	1
	Determined	[29]	1
Container Repository	Dockerhub	[37], [9]	2
	Elastic Container Registry	[24]	1
	Gitlab	[29]	1
Container Orchestrator	Kubernetes	[9], [17], [18], [34], [35], [26], [27], [41], [29]	9
	Elastic Container Service	[24]	1
Model Orchestrator	Kubeflow	[9], [31], [32], [34], [35], [27], [29]	7
	Airflow	[10], [22]	2
	Azure ML	[11]	1
	Jenkins	[33], [36]	2
	Hopsworks	[28]	1
	Scanflow-k8s	[17]	1
	AWS Sage Maker	[18]	1
Monitoring	Evidently AI	[14]	1
	Prometheus & Grafana	[17], [19], [35], [25]	4
	Neptune AI	[39]	1
Contenerization	Docker	[11], [34], [35], [25], [41], [39], [29], [36]	8
Queue	Kafka	[17], [18]	2

number of tool implementations is lower than the number of occurrences of their classes because the authors often did not indicate the name of the tool, but only their class. For example, they used a code repository without revealing a specific implementation.

The most numerous group of MLOps tools is the Model Orchestrator. It is also the 'heart' of MLOps solutions, which orchestrate the activation of all other elements of the architecture. In this class of tools, the authors list 7 solutions that were used: Kubeflow, Airflow, Azure ML, Jenkins, HopsWorks, Scanflow-k8s and AWS Sage Maker. Kubeflow is by far, the most popular among them. It is an orchestrator created and developed by Google, which can be found both on the Google Cloud Platform and on premise solutions. This is a solution that allows you to execute MLOps pipelines on distributed Kubernetes resources.

For Kubeflow to function properly, Kubernetes (9 instances) is required, which was the most popular Container Orchestration solution. It allows you to run containers and manage them in terms of performance, reliability, networking, etc. Kubernetes is also very popular in DevOps, so it is not surprising that they are widely used in the MLOps area. Docker was most often used as the implementation of the container solution, and Dockerhub was the repository.

The dominant implementation of the Model Repository was MLflow (8 instances), which allows for model management, storing files with trained models, downloading them and saving metadata related to them.

Although the Feature Store class was popular, only one implementation of this class was presented in the literature in the form of Feast. Many works discuss the available solutions of this class. Thus, while the Feature Store concept is widely recognized, its practical application remains limited. This may be due to the additional effort required to get data for the models, which is not justified by the benefits of using the Feature Store. An alternative to using this class of solutions is to use a regular database and download data directly from it. The most commonly used database is PostgreSQL (3 instances).

In monitoring, there are four solutions that are not mutually exclusive. These include Evidently AI, Prometheus, Grafana and Neptune AI. The functions of the latter also allow it to be classified as a Model Repository class. Prometheus and Grafana are also popular solutions in DevOps and allow monitoring of metrics and logs provided by running containers. What deserves special mention is Evidently AI, which is a solution dedicated to the needs of machine learning. It allows the calculation of many metrics that support the assessment of the quality and variability of data and model results during its operation. It also has ready-made dashboards that can be run in Grafana.

The next classes of tools are Code Repository and CI/CD. In the first category, the most popular is Github (3 occurrences), and in the second category, Jenkins (4 occurrences). Jenkins is still the most popular tool in this category, but this is mainly because of Jenkins' long and rich history, which has

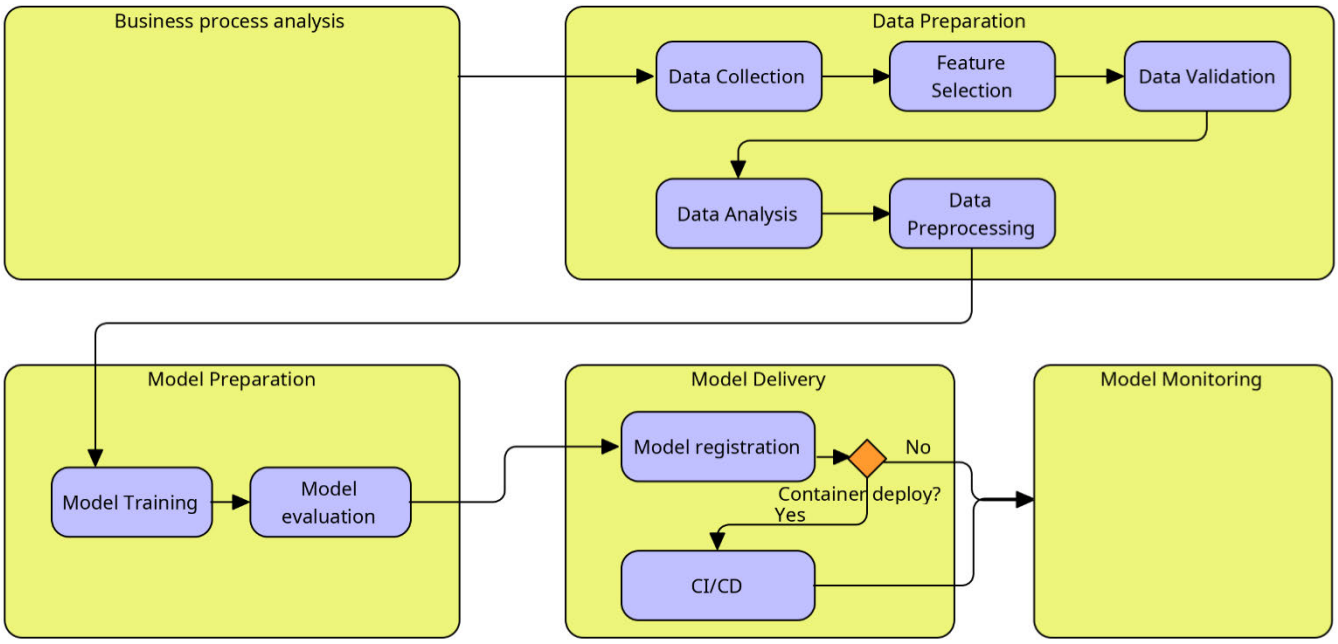

FIGURE 5. MLOps process.

TABLE 3. References to MLOps process steps in literature.

Stage	Step	Related Papers	Total num.
Business process analysis		[42], [43], [44], [26]	4
Data preparation	Data collection	[45], [21], [44], [23], [38], [46], [47], [48]	8
	Feature selection	[21], [23], [46], [36], [48]	5
	Data validation	[36]	1
	Data analysis	[45], [46], [26], [36]	4
	Data preprocessing	[45], [38], [46], [48]	4
Data preparation	Model training	[42], [45], [21], [43], [44], [23], [38], [46], [26], [47], [36], [48]	12
	Model evaluation	[42], [45], [43], [44], [38], [46], [26], [47], [36], [48]	10
Model delivery	Model registration	[45], [21], [23], [26], [36]	5
	CI/CD	[42], [45], [21], [23], [38], [46], [26], [36], [48]	9
Model Monitoring		[42], [45], [21], [44], [23], [46], [47], [36], [48]	9

led to its dominance in this area. Recently, however, solutions that are easier to maintain, such as Github Actions and GitLab CI, have been gaining popularity. The only implementation in queue management that we have encountered is Kafka.

Within the analyzed materials, we identified two solutions belonging to the Distributed Computation class: Flower and Determined. Both frameworks enable the distribution of tasks within the model training process, particularly for hyperparameter search.

C. AD RQ3: MLOps PROCESS

The process of creating and maintaining models comprises different steps depending on the source. These differences have two causes. The first is differing interpretations of the process, leading to variations in how steps are defined and grouped, even though the core actions remain consistent. Here, the difference is only in the naming and grouping of activities into steps. Differences also arise from model architectures and usage methods. For example, model scoring can be performed cyclically within the pipeline of the Model Orchestrator or online using the REST API served by container. This results in significantly different actions required to use the model.

All the steps that we extracted from the literature are included in the Fig. 5, and references to the descriptions of the steps are provided in the Table 3. We have divided the steps into four stages. The first stage involves the analysis of the business problem and it was not further developed in

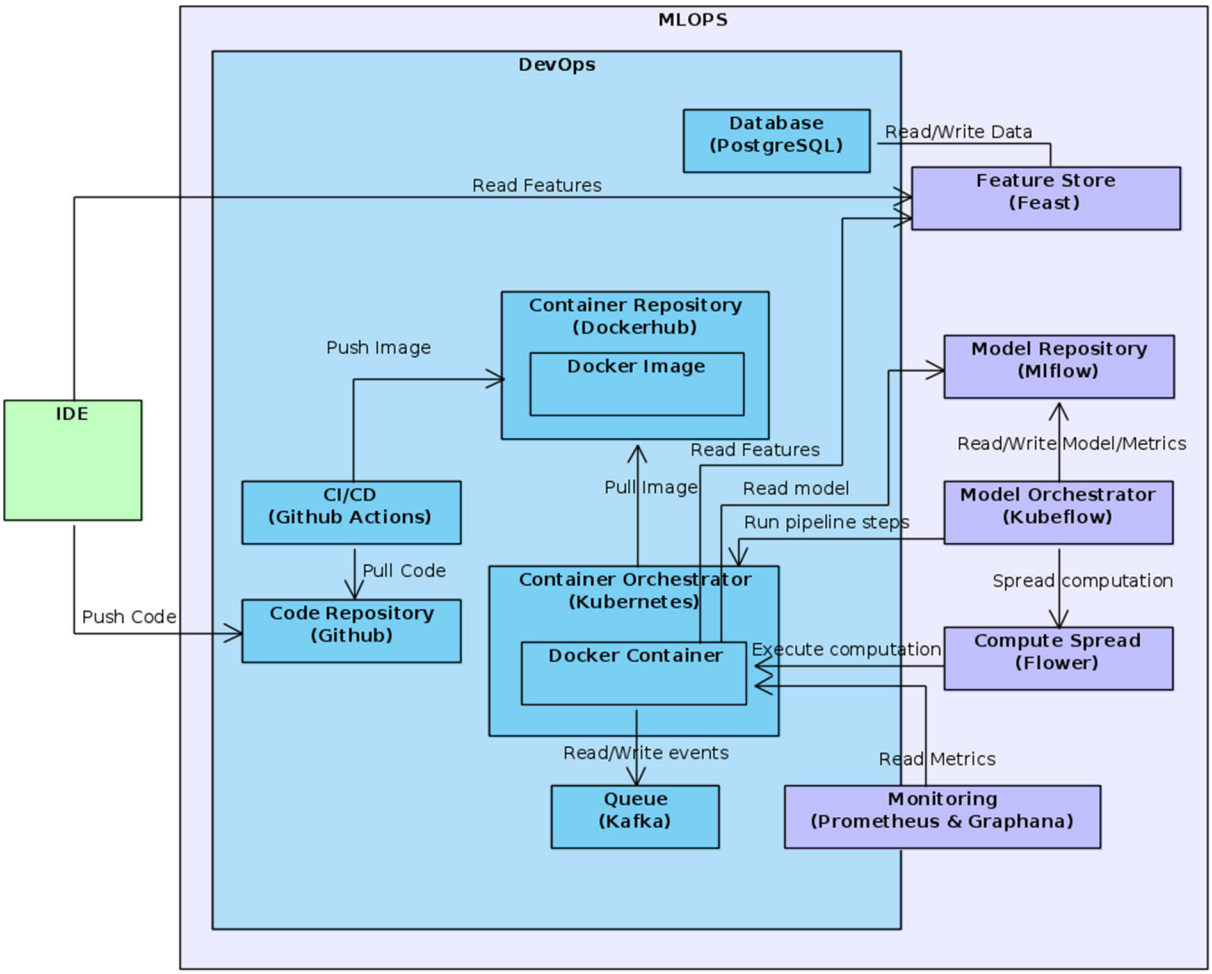

FIGURE 6. MLOps reference architecture.

the analyzed literature because it is tangential but outside the scope of MLOps.

The name we chose for the second stage is 'Data Preparation'. It includes the steps: Data collection, Feature selection, Data validation, Data analysis, Data preprocessing. The names of the steps we mention are not exactly the same in the cited sources. For example, you may encounter the names feature selection and feature engineering. Since these steps involve similar activities, we decided on a common name. Other steps, which may appear under different names in different sources, follow the same principle. The feature selection step aims to extract features from the data that will be used in subsequent steps of model preparation. The Data validation step covers slightly broader activities than those typically implemented in software. Namely, in addition to verifying data structures and correctness, their completeness, timeliness, consistency, reliability, etc. are also checked. Data analysis includes detecting relationships between data and assessing the possibility of their use in the model. The data preprocessing step includes transforming data into the forms required by the trained model, such as variable coding and normalization.

The next stage is Model Preparation. It includes Model training and Model evaluation. Model training can be performed directly by the orchestrator (e.g. Kubeflow), or the orchestrator can ask another tool to distribute the computation, or the scattering can be performed in parallel at both levels. After the model is trained, it is evaluated. Model evaluation parameters are calculated, depending on its type, which will ultimately be sent to the repository together with the model.

After the model is prepared, it is deployed. Depending on how the model is used, this may be a simpler or more complex operation. A simpler variant is when the implementation involves registering a new model in the registry and then marking it as production. In this variant, the software that uses the model will download its new version from the repository. In particular, it may be a pipeline managed by a

model orchestrator or a service that uses models as part of a provided REST API. A more complex variant of this process involves implementing a microservice that exposes the model via an API. In this variant, in addition to the previously mentioned steps, CI/CD processes must occur, the scope of which is analogous to traditional DevOps process.

After model deployment, the last step is to monitor the solution. It includes both data monitoring and metrics that evaluate the quality of model operation. Monitoring data is especially important when data drifts occur. Regardless of the monitored area, the detection of anomalies may cause the decision to start the entire process from the beginning.

D. AD RQ4: MLOps EFFICIENCY METRICS

Unfortunately, we did not find an answer to Research Question 4 in the analyzed literature. None of the reviewed articles presented metrics that could measure the effectiveness of MLOps implementation in an organization. However, we found two publications that presented methods for assessing the maturity of MLOps processes in an organization, namely: [2], [49]. The content of these articles is closest to the answer to the question we asked, but they do not answer it.

In [2], the authors list the following stages of MLOps Maturity Model: Ad hoc, DataOps, Manual MLOps, Automated MLOps, Kaizen. For each stage, they describe the criteria that are necessary to achieve it. The text [49] presents methods for evaluating the capability to produce models. They list the different components of the assessment in the areas of: Data Integration, Data Preparation, Modeling, Deployment.

We identified several potential reasons for the absence of publications addressing metrics. The most likely reason is the short history of MLOps implementations. In order to develop metrics, and above all to discuss market standards for these metrics, there must be a sufficient base for their comparison. Therefore, MLOps would have to have a more established form. This is especially true for stabilizing the scope of MLOps, because different authors interpret it differently. There is a rich and stable base of metrics related to classic software development. This may satisfy some needs for metrics in the MLOps area. The last reason we suspect is that organizations have developed individual metrics to evaluate MLOps implementations, but they have not been published. This leads us to a proposal for further research in this area. Future research should analyze MLOps implementations in organizations to identify existing metrics and establish a standardized framework. In the absence of such metrics, we would propose metrics based on the analysis of MLOps processes.

V. CONCLUSION AND FUTURE WORK

MLOps is a young and growing field of machine learning. In the reviewed articles, one could find many architectural proposals, used tools and processes. Despite the variety of approaches, certain foundational elements and leading solutions can be identified as the core of MLOps. These solutions can be divided into two parts: basic DevOps and dedicated MLOps. The first category includes CI/CD tool classes, Code Repository, Container Orchestrator. The second category includes primarily the following: Model Orchestrator, Model Repository and Feature Store. A summary of MLOps tool classes, together with the most popular solutions, is presented in Fig. 6. It is also the MLOps reference architecture, which can be referred to when building dedicated solutions. We also defined the model production process in the MLOps environment. However, market standards for some steps, such as model monitoring, have not yet been developed. Finally, the absence of publications detailing metrics for evaluating MLOps implementations was unexpected. Perhaps this is due to the immaturity of this area and measurable indicators will be developed to assess the quality of the implemented MLOps environment and compare these indicators between companies.

In the future, we plan to work in three areas. First, we want to address the problem of limited answers to the question about metrics. We want to conduct research in private companies and prepare a proposal for metrics for MLOps implementations. Second, we want to continuously update the MLOps architecture and publish when significant changes to this work occur. Third, we want to expand the sources of information about MLOps implementations to include private companies that do not publish their results but agree to share them with us.

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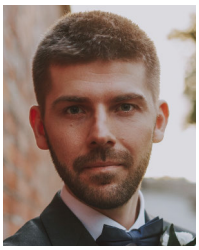

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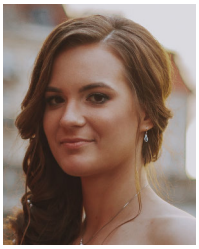

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